CEED 2013 COMPLETE SOLUTION

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CEED 2013 Part A

Fifty Questions,

Correct answer carries + 2.0 marks, Wrong answer carries- 0.5 marks.

Question No. 1 Ans 3

The resolution of a full HD is

Options :

1. 800 X 600 pixels
2. 1024 X 768 pixels
3. 1920 X 1080 pixels
4. 768 X 576 pixels

Standard display resolutions

Name	Resolution in pixels
High Definition (HD)	1280x720
Full HD	1920x1080
2K, Quad HD	2560x1440
4K, Ultra HD	3840x2160

Question No. 2 Ans 2

The frame rate of PAL video is

Options :

1. 24 fps
2. 25 fps
3. 30 fps
4. 8 fps

Phase Alternating Line (PAL) is a colour encoding system for analogue television. It was one of three major analogue colour television standards, the others being NTSC and SECAM. In most countries it was broadcast at 625 lines, 50 fields (25 frames) per second, and associated with CCIR analogue broadcast television systems B, D, G, H, I or K.

Important Computer Abbreviations

CPU=Central Processing Unit
RAM = Random Access Memory
ROM = Read Only Memory
PROM = Programmable Read Only Memory
EPROM = Erasable PROM
EEPROM = Electrically EPROM
HDD = Hard Disk Drive
FDD = Floppy Disk Drive
KBD = KeyBoard
I/O = Input & Output
CD = Compact Disk
DVD = Digital Video Disk
SMPS = Switch Mode Power Supply
POST = Power ON Self Test
BIOS = Basic Input Output System
VDU = Visible Display Unit
LED = Light Embedded Diode
LCD = Liquid Crystal Display
USB = Universal Serial Bus
VGA = Video/Visual Graphic Adapter

Question No. 3 Ans 3

HDMI stands for

Options :

1. High Density Material Interface
2. High Dimension Material Interface
3. High Definition Multimedia Interface
4. High Definition Material Interface

Question No. 4 Ans 1

The Director of the recent animation film "Arjun the warrior prince" is

Options :

1. Arnab Chaudhuri
2. Anurag Kashyap
3. Arnab De
4. Karan Johar

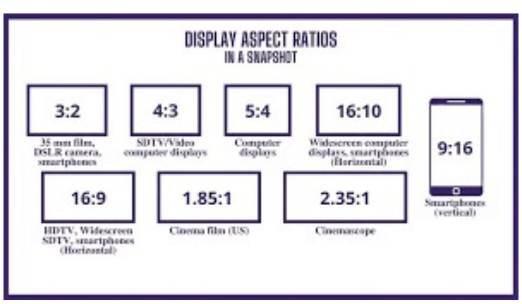
The Warrior Prince is a 2012 Indian animated action film, directed by Arnab Chaudhuri, written by Rajesh Devraj, and produced by Ronnie Screwvala and Siddharth Roy Kapur under UTV Motion Pictures and Walt Disney Pictures.

Question No. 5 Ans 3

The ratio of horizontal dimension to vertical dimension for "wide screen" format is likely to be:

Options :

1. 4:3
2. 4:4
3. 16:9
4. 16:12



Question No. 6 Ans 1

The character belongs to which comic strip ?

Options :

1. Garfield
2. Tommy
3. Black stripe
4. Harfield

Question No. 7 Ans 3

The partial view of a common car is shown below. Identify the car.

Options :

1. New Wagon R
2. Zen Estillo
3. Santro
4. Spark

Question No. 8 Ans 2

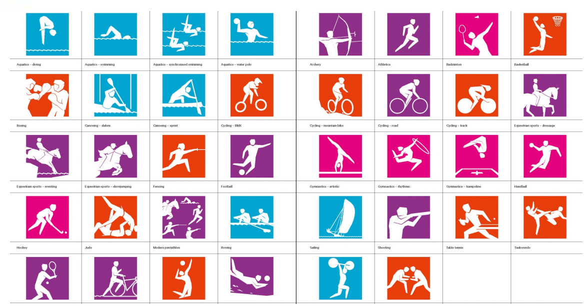
Several Fonts are shown below. Identify the one that can be used for Invitation Cards for a joyous occasion.

Options :

1. A
2. B
3. C
4. D

Question No. 9 Ans 3

Identify the Pictogram for swimming in London Olympics 2012



Options :

1. A
2. B
3. C
4. D

Question No. 10 Ans 2

A reflected image is shown below, Identify the reflective surface.

Options :

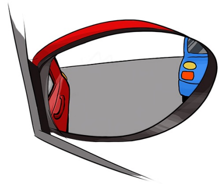
1. Concave surface
2. Convex surface
3. Cylindrical surface
4. Conical surface

Question No. 11 Ans 2

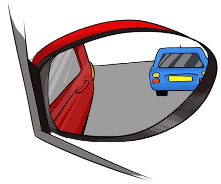
The Rear view mirror of a car is

Options :

1. Concave Mirror
2. Convex mirror
3. Plain mirror
4. None of these



FLAT MIRROR



CONVEX MIRROR

Question No. 12 Ans 4

Mary Kom is associated with

Options :

1. Wrestling
2. Kung Fu
3. Karate
4. Boxing

Question No. 13 Ans 1

Among the objects that have equal length, breadth and height, the least surface area is of

Options :

1. Sphere
2. Cube
3. Cone
4. Cylinder

Question No. 14 Ans 2

CYMK stands for

Options :

1. Coronary Multiple Yield Ketosis
2. Cyan, Magenta, Yelow, Black
3. Crimson, Magenta, Yelow, Kala
4. Cartoon Memory Yielding Knowledge

Question No. 15 Ans 3

In a computer, what will be the resultant colour if you use the following combination: R = 100, G = 100, B = 100.

Options :

1. Black
2. White
3. Gray
4. Yellow

Question No. 16 Ans 4

Identify the animal from the skin texture

Options :

1. Giraffe
2. Tiger
3. Zebra
4. Cheetah

Question No. 17 Ans 1

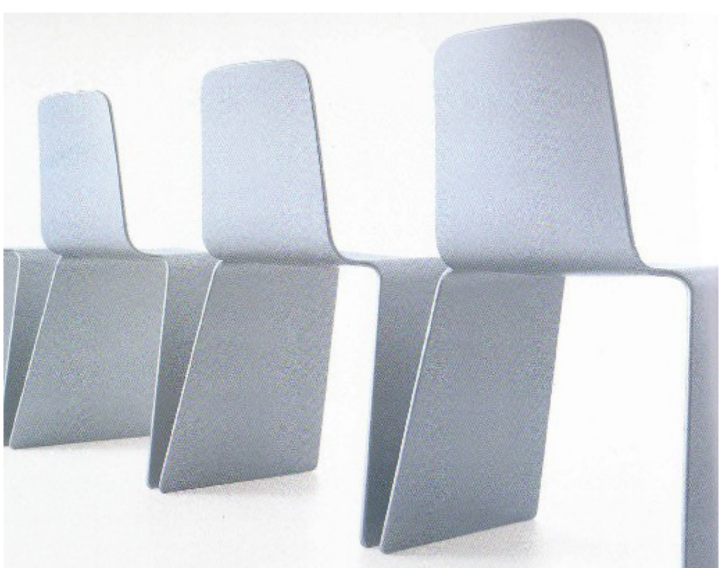
Identify the object shown in the image

Options :

1. Tibetan prayer wheel
2. Wind Chime
3. Chinese musical instrument
4. Bell for monkey dance

Question No. 18 Ans 2

An image of a chair is shown below. Identify the material most suited for it's manufacture.



Options :

1. Steel
2. Plastic
3. Laminated wood
4. Cast iron

Question No. 19 Ans 3

Identify the state to which this head gear belongs to

Options :
1. Bihar
2. Kerala
3. Rajasthan
4. West Bengal

Question No. 20 Ans 4

Blue Tooth stands for

Options :
1. Damaged tooth turned blue
2. Tooth of Blue Whale
3. Wireless navigation system
4. Wireless communication system

Question No. 21 Ans 1

Identify the type of lens used to shoot the photograph shown here.

Options :
1. Fish eye lens
2. Wide angle lens
3. Tele lens
4. Zoom lens

Focal Length	Type of Lens	What is it used for?
4mm - 14mm	Fisheye	Abstract, creative
14mm - 35mm	Wide angle	Landscape, architecture
35mm - 85mm	Standard	Street, travel, portrait
85mm - 135mm	Short telephoto	Street photography and portraits
135mm+	Medium telephoto	Sports, wildlife, action
300mm+	Super telephoto	Sports from a distance, nature and astronomy
35mm - 200mm	Macro	Close-up shots

Question No. 22 Ans 2

Identify the product in the photograph.

Options :

1. Ironing box
2. Carpet Cutter
3. Gardening tool
4. Surgical tool

Question No. 23 Ans 4

Identify the Musical instrument shown below

Options :

1. Sitar
2. Tanpura
3. Sarod
4. Rudra Veena

Question No. 24 Ans C

Identify the oldest camera from the picture

A B ~~C~~ D

Options :

1. A
2. B
3. ✓ C
4. D

Question No. 25 Ans 2

A flat belt pulley has a circumference of 40 cms and turns at 60 rpm. It rotates another pulley of 1.2 meters circumference through a belt having 7.2 meters in length. If driven pulley has revolved 600 times, what is the time that the driver pulley rotated?

Options :

1. 20 minutes
2. ✓ 30 minutes
3. 40 minutes
4. 120 minutes

$$\frac{N_1}{N_2} = \frac{D_2}{D_1} = \frac{2\pi R_2}{2\pi R_1}$$

$$\frac{\cancel{60}}{x} = \frac{\cancel{120}}{\cancel{40}} \qquad N_2 = 20\ rpm$$

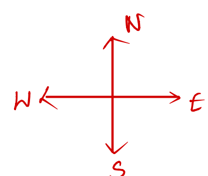
$\therefore$ 1 min $\longrightarrow$ 20

x min $\longrightarrow$ 600

$x = 30$ min

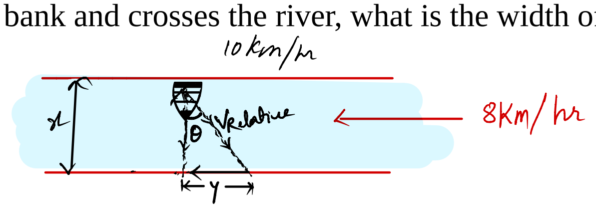
Question No. 26 Ans 1

A boat is crossing a wide river flowing at 8 km/hr from east to west. The landing port is exactly opposite the starting point in the north bank of the river. If the boat moves at 10 km/hr towards other bank and crosses the river, what is the width of the river?



Options :

1. ✓ 6 km
2. 8 km
3. 10 km
4. 1.8 km



$V_{Resultant\ boat} = 10\ Km/hr$

$10 = \sqrt{8^2 + V_B^2}$

$\therefore$

$100 - 64 = V_B^2$

$V_B^2 = 36 \Rightarrow V_B = 6\ Km/hr$

If assume Time to be 1 hr then width = 6 km

Question No. 27 Ans 2

Local time of a place is related to the value of.

Options :

1. Latitude of the place
2. ✓ Longitude of the place
3. Height of the place from mean sea level
4. Distance from the equator

The rate of the time at which the sun traverses over certain degrees of longitudes is used to determine the local time of an area with respect to the time at the Prime Meridian (0°Longitude).

latitude and longitude, in cartography, a coordinate system used to determine and describe the position of any place on Earth's surface. Latitude is a measurement of a location north or south of the Equator. In contrast, longitude is a measurement of location east or west of the prime meridian at Greenwich (an imaginary north-south line that passes through both geographic poles and Greenwich, London, England, U.K.). Latitude and longitude together can describe the exact location of any place on Earth.

Question No. 28 Ans 1

Find the odd object out.

Options :

~~1.~~ Sphere
2. Tetrahedron
3. Cube
4. Cylinder

Question No. 29 Ans 2

The front and rear wheels of a tractor have different diameters in the proportion of 1:2 ratio. What will be the speed of the front wheel, if the rear wheel moves at 40 km/hr on road ?

Options :

1. 80 km/hour
~~2.~~ 40 km/hour
3. 60 km/hour
4. 120 km/hour

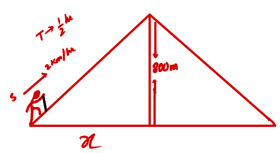
what ever be the proportion as long as it is on road and moving The velocity will be same as rear wheel

Question No. 30 Ans 4

A pilgrim is climbing to a temple at top of a conical hill . The hill has equal slope from bottom to top and is 800 meters high from it's base. The pilgrim covered the climb at the rate of 2 km/hour in half an hour. What will be the diameter of the base of the hill ?

Options :

1. 1800 meters
2. 1600 meters
3. 600 meters
~~4.~~ 1200 meters



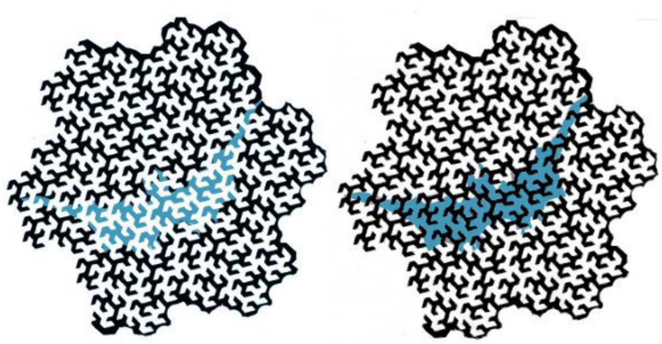
$S = \frac{D}{T}$

$D = S \times T$
$= 2 \times \frac{1}{2}$
$= 1\,km = 1000\,m$

$(1000)^2 = (800)^2 + x^2$ (phythogores theorem)
$(1000)^2 - (800)^2 = x^2$
$(1000 - 800)(1000 + 800) = x^2$
$200 \times 1800 = x^2$
$600 = x$
$\therefore$ Diameter $= 2x$
$= 1200$

Question No. 31 Ans 4

Observe the two fractal patterns given below and the patch of color in them.

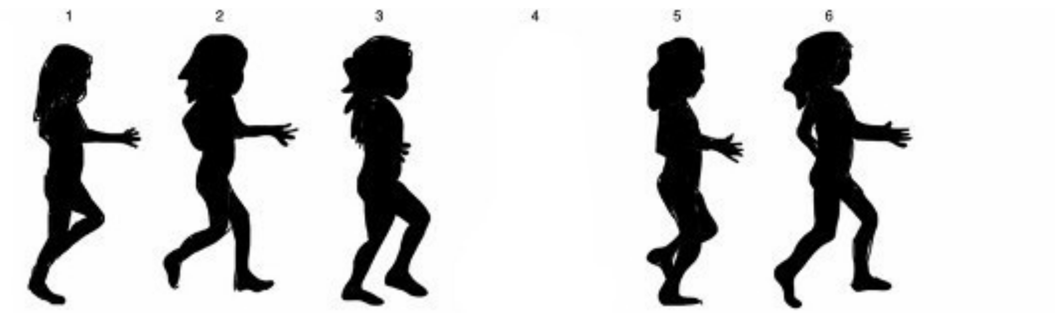


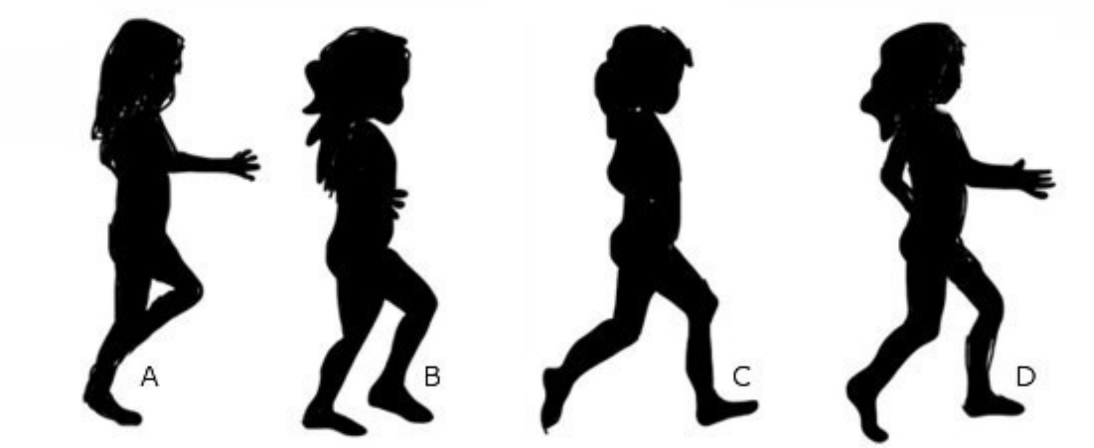
Options :

1. The patch of color in the left hand pattern is darker
2. The patch of color in the right hand pattern is lighter
3. Both the patterns have patches of color with the same tone
~~4.~~ None of the above options

Question No. 32 Ans 3

The sequence of numbered images depicting a walk cycle below, image number 4 is missing. Of the four options given, select the correct image.



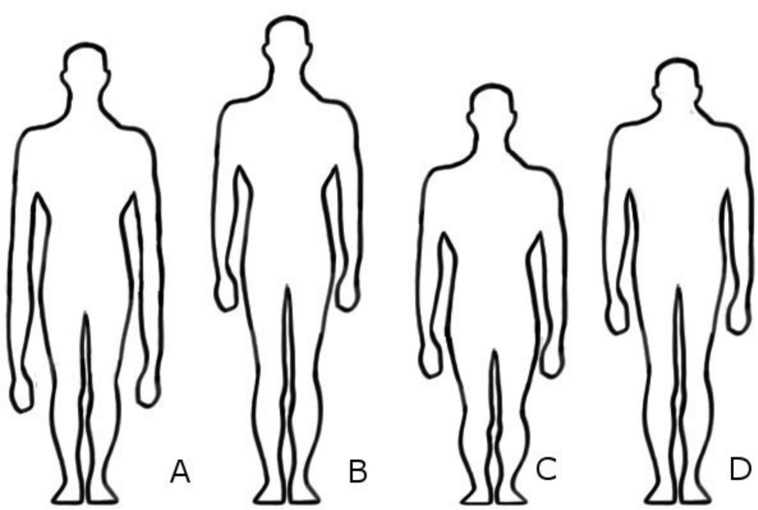


Options :

1. A
2. B
3. C
4. D

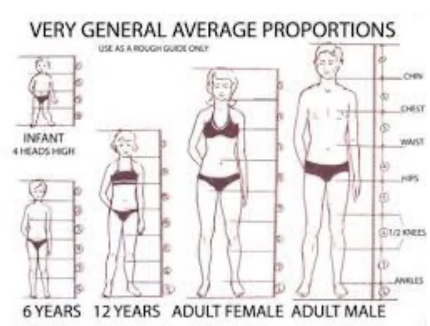
Question No. 33 Ans 2

The correct proportion of the human figure is:



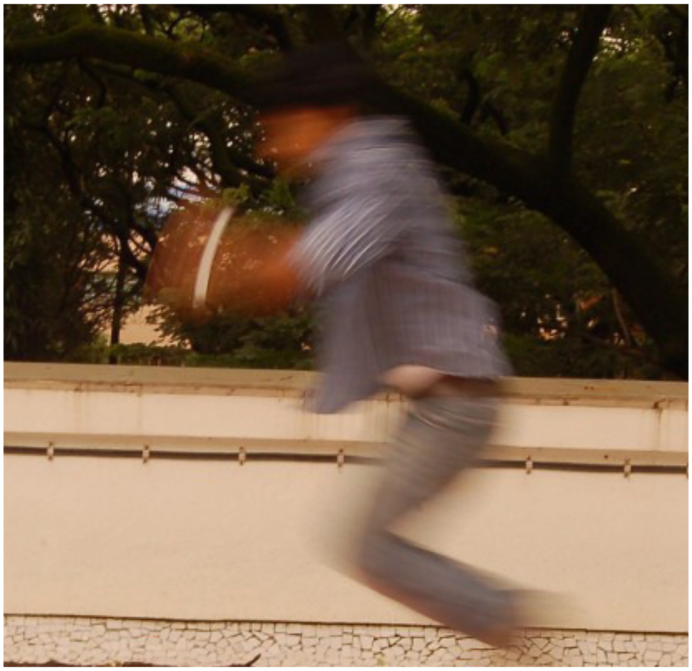
Options :

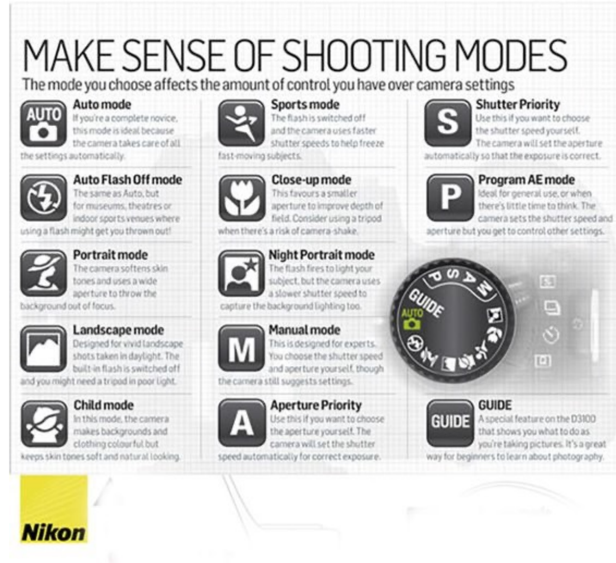
1. A
2. B
3. C
4. D

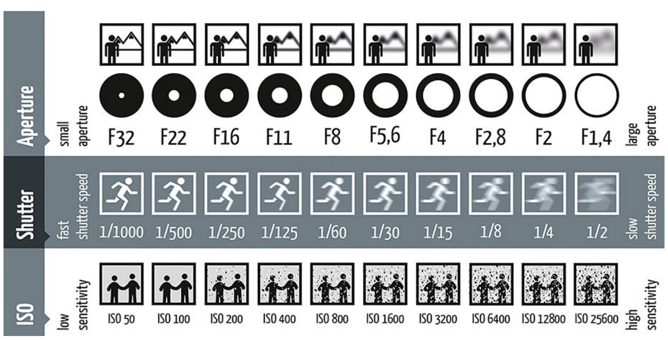


Question No. 34 Ans 1

Observe the photograph and choose the correct option:







Options :

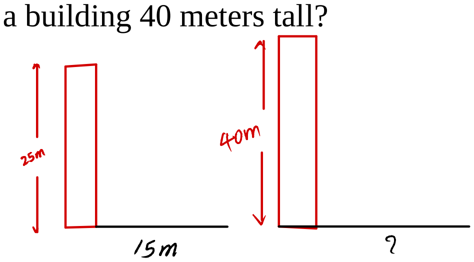
1. The shutter speed setting was low
2. The shutter speed setting was high
3. There was a camera shake while clicking the picture
4. There was a mistake while saving the image

Question No. 35 Ans 2

A building of 25 meters casts a shadow 15 meters long on the ground. What will be the length of shadow cast by a building 40 meters tall?

Options :

1. 20 meters
2. 24 meters
3. 26 meters
4. 30 meters



$\frac{25}{15} = \frac{40}{x}$

$x = \frac{\cancel{15}^{3} \times \cancel{40}^{8}}{\cancel{25}_{5}}$

$x = 24$

Made with Goodnotes

Question No. 36 Ans 2

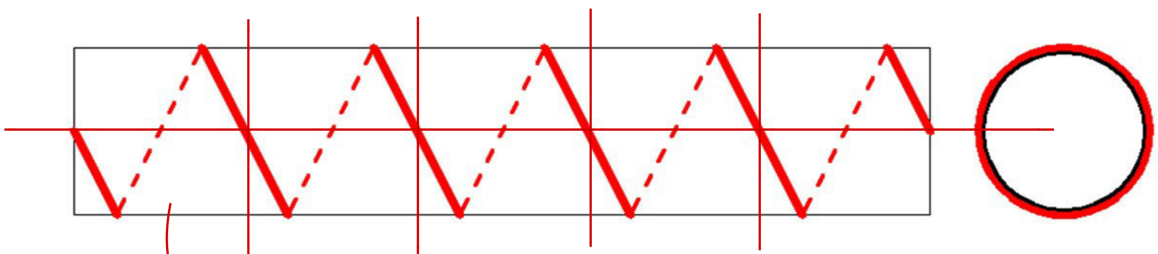
A number of children are standing evenly spaced around a circle. The third child is directly opposite to 28th child. How many children are there altogether?

Options :

1. 48
2. 50
3. 54
4. 56

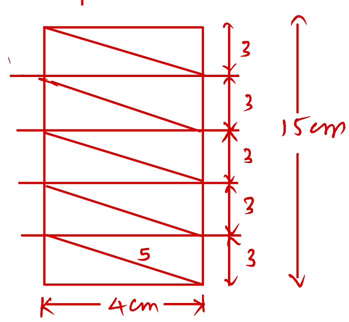
Question No. 37 Ans 3

A cylinder 15 cms long has a circumference of 4 cms. A string makes exactly 5 turns around the cylinder, while it's two ends touch the cylinder top and bottom. How long is the string?



Options :

1. 20 cms
2. 22 cms
3. 25 cms
4. 28 cms



$\therefore$ 5 × 5turns = 25cm

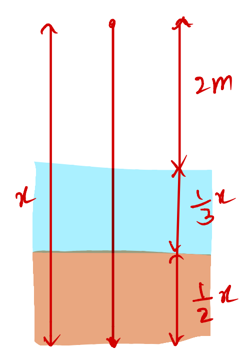
Ans 2

Question No. 38

There is a pole in a lake, one half of the pole is buried in soil, another one third is submerged in water, and 2 meters is out of the water. What is the total length of the pole?

Options :

1. 10 meters
2. 12 meters
3. 14 meters
4. 18 meters



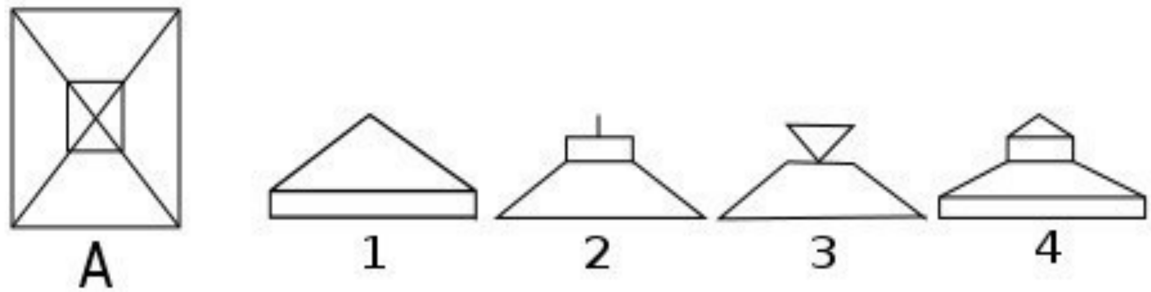
$2 + \frac{1}{3}x + \frac{1}{2}x = x$

$12 + 2x + 3x = 6x$

$12 = x$

Question No. 39 Ans 4

The figure "A" here is a plan view drawing of one of the objects shown in the right. Identify the correct object whose plan view drawing is shown.



Options :

1. 1
2. 2
3. 3
4. ~~4.~~ 4

Question No. 40 Ans 1

Identify the aircraft which uses the latest technology

A B C D

Options :

1. ~~1.~~ A
2. B
3. C
4. D

Question No. 41 Ans 2

Identify the foreign car manufacturer that does not have a manufacturing facility in India?

Options :

1. Nissan
2. ~~2.~~ Citreon
3. Renault
4. Skoda

Question No. 42 Ans 2

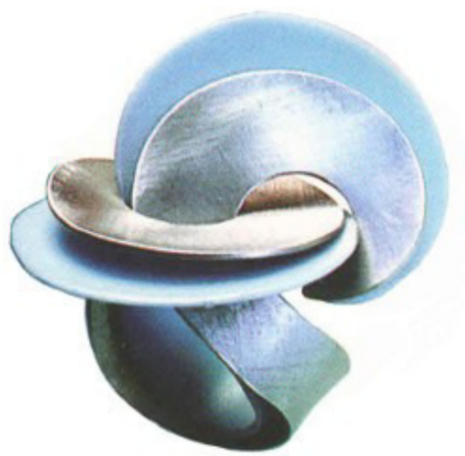
The following product is used as?

Options :

1. Packaging
2. Pillow
3. Fruit basket
4. Paper weight

Question No. 43 Ans 1

The following products are used as?



Options :

1. Hair Clip
2. Key Ring
3. Christmas Decoration
4. Finger Ring

Question No. 44 Ans 2

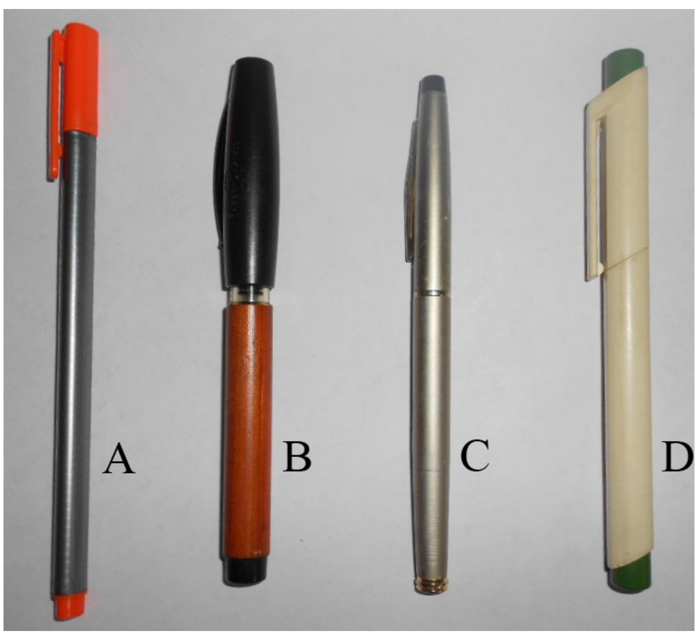
Which of these torch lights is suitable for a security guard?

Options :

1. A
2. ~~2.~~ B
3. C
4. D

Question No. 45 Ans 3

Which of the following pens is sporty?



Options :

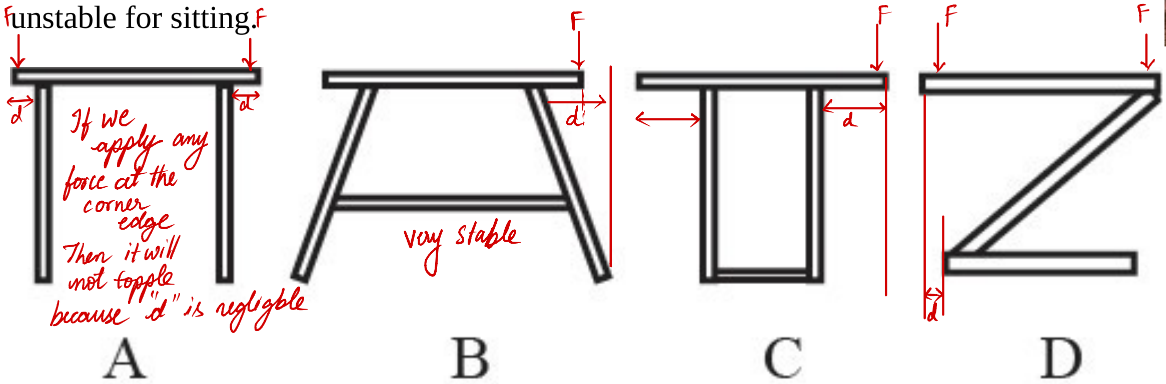
1. A
2. B
3. C
4. D

Torque = force × perpendicular distance

$= F \times d$

Question No. 46 Ans

The following are drawings of different stools. Indicate which one of them is most unstable for sitting.



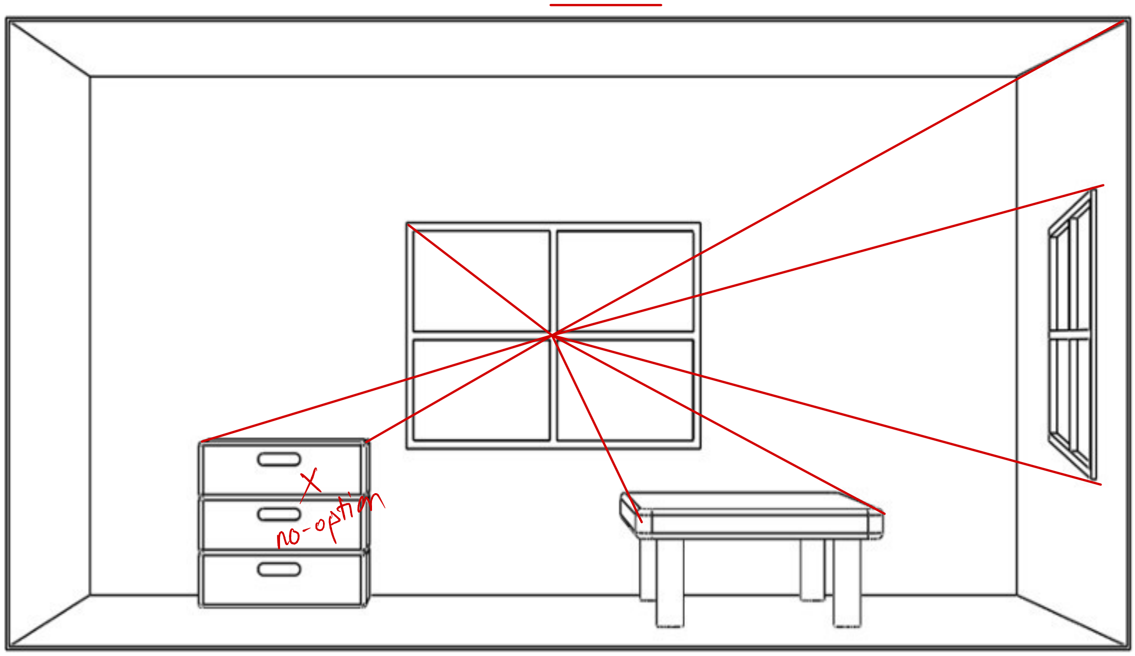
Options :

1. A
2. B
3. C
4. D

∴ from the given option most unstable is C

Question No. 47 Ans

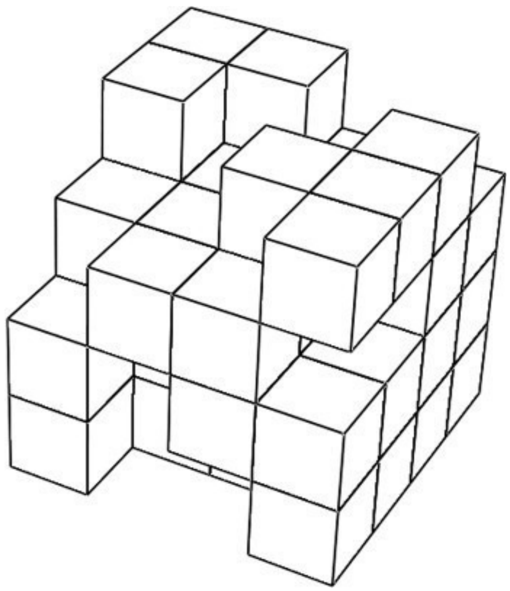
Which object or surface in the given figure is/are not in perspective?



Options :

1. Walls
2. Table
3. Window on the back wall
4. Window on the side wall

Question No. 48
The figure shown below is a composition of cubes of four by four by four matrix. Some of the cubes are removed, but no cubes those are not visible are removed. How many cubes are there in the whole composition?



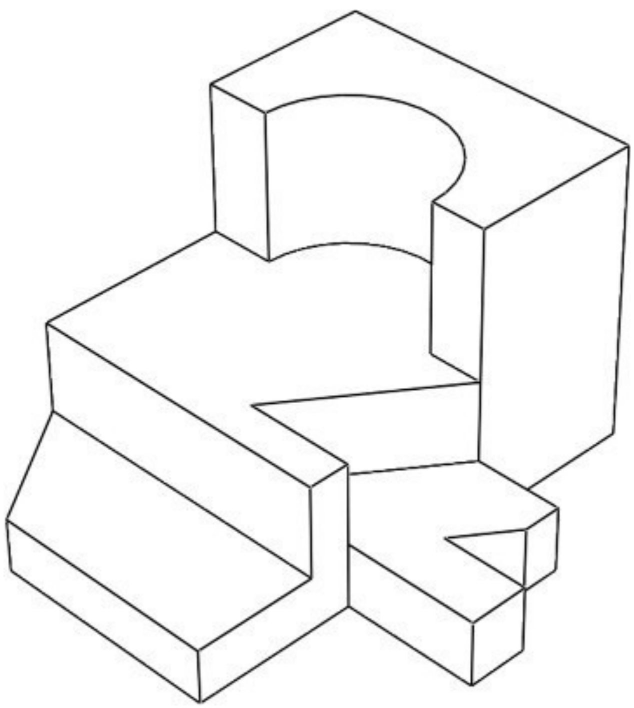
Options :
1. 45
2. 47
3. 49
4. 51

Total no. of cubes is $4 \times 4 \times 4$ is 64

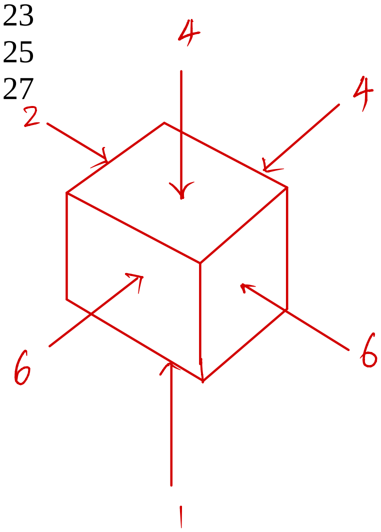
missing cubes
1st row – 9
2nd row – 3
3rd row – 1
4th row – 2
Total – 15

$\therefore 64 - 15 = 49$

Question No. 49
Count the number of surfaces in the shown solid.

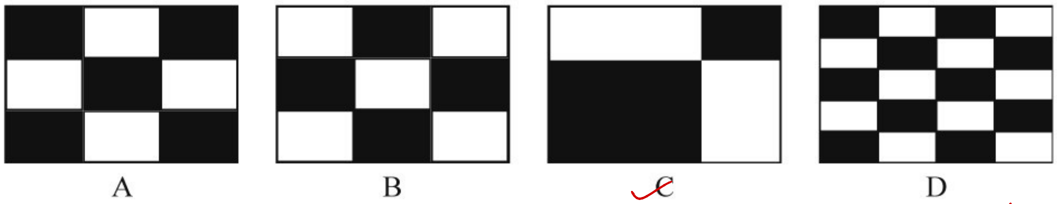


Options :
1. 21
2. 23
3. 25
4. 27



Question No. 50 Ans C

Study the visuals shown here. Select one which you find aesthetically pleasing.



Options :

option A, B, D are more or less looking same but C is quite aesthetically pleasing

1. A
2. B
3. C
4. D

CEED 2013 Part B Mandatory

Following three questions are mandatory

Question No. 1 **Marks 20**

Create a freehand sketch of an apple and a bunch of six bananas placed on a dinner plate.

The final drawing is to be done freehand using only pencil. You may use shading to create a realistic three dimensional feel.

Do not use any colour.
Do not use any drawing instrument like ruler, set square, compass etc.

Evaluation criteria:
Perspective
Light, shade and shadow
Quality of lines
Composition
Three Dimensional Quality

Question No. 2 **Marks 15**

You are given a special magic liquid. You can make anything invisible by spraying this liquid on it. Describe five instances where you will use this liquid in your daily life.

Use a sketch and a brief note (less than 20 words) to describe each answer.

Evaluation criteria:
Originality of ideas.
Ability to imagine.
Sense of humor

Question No. 3 **Marks 15**

Below is an image of a college canteen in India. Identify five problems in this scenario. Briefly describe the problems (in less than 20 words each).

Evaluation criteria:
Identification of unique problem.
Ability to understand the problems.
Design Sensitivity.

Attempt any one of the following three optional questions.

CEED 2013 Part B Optional (Product Design)

Marks 50

In case of day to day activity in a household, one tedious activity is mopping of the floors. A normal mop cannot reach all the crevices and under the furniture.
Design a manually operated floor mop for domestic use.

1. Identify at least five distinct factors essential for designing a mop that can be used for domestic purpose.
2. Generate three distinct concepts through pencil sketches based on the factors identified above.
   Present these pencil sketches along with brief notes (not more than 15 words).
3. Create your final concept and present this through drawings, showing the overal form and design of it's features. Indicate materials to be used for various parts of the mop.

Evaluation criteria:
Identification of distinct factors.
Originality of your alternate concepts.
Appropriateness of your final solution.
Appropriateness of choice of materials.
Quality of presentation

CEED 2013 Part B Optional (Visual Communication)

Marks 50

The following instructions are required to be printed on a medicine packet. Represent visually the given instructions without using any text.

1. One tablet before breakfast at 7:00 AM.
2. One tablet after lunch at 2:00 PM.
3. One and half tablets after dinner at 9:00 PM.
4. Not to be taken by pregnant women

Size of the medicine packet is as follows:
Height: 50mm, Width 200mm, Depth: 50mm.
Your final artwork (total ALL four visuals) should fit into the fol owing size: Height: 40mm, Width 160mm
Explain your design solution in not more than 20 words.

Evaluation criteria:
Ability to communicate clearly using visuals without use of text.
Clarity of visuals.
Quality of renderings.
Overall effectiveness of final design solution.

CEED 2013 Part B Optional (Animation Design)

Marks 50

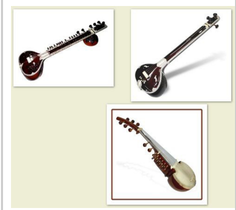
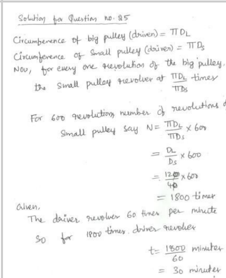
The crow is the most common bird that we get to see in India. One cannot miss noticing the innumerable deep rooted associations that the crow has in Indian society.
Imagine that you are a crow and narrate your autobiography.

1. Write your story in 5 sentences.
2. Make 3 sketches of the crow in your story in various poses/moods/expressions.
3. Design a storyboard to describe your story in 8 frames.

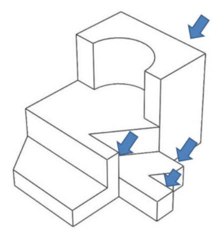
Evaluation criteria:
Ability to design an interesting character and story.
Quality and finish of drawings.
Originality of the story.

CEED 2013

Q.No	Ans	Explanation	Ans image	Learn about more
1	3	Full HD measures 1920X1080 pixel (2.1 MP) with widescreen aspect ratio of 16:9		http://stuffyoulook.blogspot.com/2013/10/video-and-color-technology.html
2	2	PAL is "Phase Alternating Line"; is a color encoding system for television system used in broadcast television systems in countries like Europe, Australia, middle east, Asia broadcasting at 576 vertical lines (720 X 576) thus counting to 25 frames per second (fps). conventional television screens are made up of horizontal lines, while computer monitors consists of a series of horizontal and vertical pixels.		http://stuffyoulook.blogspot.com/2013/10/video-and-color-technology.html & http://www.adobe.com/devnet/flash/learning_guide/video/part06.html
3	3	HDMI is a port (compact audio cum video interface) for transferring video/audio data to a compatible monitor, projector, TV, or digital audio		
4	1			
5	3	Widescreen format is HD mode fitting to 16:9 ratio screen		
6	1	Related people in Garfield cartoon		http://stuffyoulook.blogspot.com/2013/10/Animation-study-for-Part-A.html
7	3			
8	B			
9	3			
10	2			
11	2	Using concave mirrors will invert the image, which is not feasible		
12	4	She is a famous Indian boxer and won world boxing champion five times		
13	1			
14	2	CMYK is a general color space used for commercial printing and most color computer printers. encyclopedia says that CMY can print all colors. K stands for black is required for quality printing.		http://stuffyoulook.blogspot.com/2013/10/video-and-color-technology.html
15	3	Check your Photoshop software for different color formats including CMYK		

16	4	Only cheetah have small black spots		
17	1	It's a Tibetan prayer instrument of about one feet size		
18	2	Usually plastic chairs requires cross legs for stability, But its odd !!! I found that most of the new model chairs are usually made of PLASTIC for better ergonomics		
19	3			
20	4	Bluetooth is basically meant to CONNECT several devices and is an alternative to RS-232 data cables		
21	1	A fish eye lens is an ultra wide - angled lens that produces strong distortion intended to create a wide panoramic or hemispherical image.		http://stuffyoulook.blogspot.com/2013/10/photography-for-CEED-Part-A.html
22	2			
23	4	Only Rudra veena has two equal elliptical domes attached sitar, Tanpura and Sarod were shown below		http://stuffyoulook.blogspot.com/2013/10/indian-musical-instruments.html
24	3			http://stuffyoulook.blogspot.com/2013/09/past-to-present-evolution-of-products.html
25	2	Length has nothing to do with revolutions, for every one revolution of driven (big pulley), the driver (small pulley) must have revolved thrice, thus if driven revolves 600 times, the driver should have revolved 1800 times (i.e 30 minutes) as it rotates at 60 rpm.	 Solution for Question no. 25 Circumference of big pulley (driven) = πD_L Circumference of Small pulley (driver) = πD_S Now, for every one revolution of the big pulley, the small pulley revolves at $\frac{\pi D_L}{\pi D_S}$ times For 600 revolutions number of revolutions of Small pulley say $N = \frac{\pi D_L}{\pi D_S} \times 600$ $= \frac{D_L}{D_S} \times 600$ $= \frac{120}{40} \times 600$ $= 1800$ times Given, The driver revolves 60 times per minute So for 1800 times driver revolves $t = \frac{1800}{60}$ minutes $= 30$ minutes	
26	1	To reach exactly opposite point of the bank from where the boat has started; the boat should move slant towards north east at 10kmph, and water is flowing at 8 kmph from east to west, now apply Pythagoras theorem to find the relative velocity which turns to be 6 kmph.		
27	2	A lines of longitude is also called a "Meridian", meaning middle day or Noon, and times of the day before noon were known as "ante meridian" (AM), and times of the day after it were called as "Post meridian" (PM) check more about time zones		http://www-istp.gsfc.nasa.gov/stargaze/Slatlong.htm
28	1	Sphere does not have a sharp edge; while the remaining (tetrahedron,cube and cylinder) does have edges		
29	2	It's not 80 kmph, funny right !! the front and back wheel will be traveling same distance in a unit time regardless of the wheel size. This question checks your thinking ability.		

30	4	Apply Pythagoras theorem, here the vertical line (in mathematics we call opp. side) is 800 m and the slant line (called hypotenuse) is 1000 m, thus resulting to 600 m as radius, thus dia probably assumes 1200 m. Slant line - 1000m ? This is simple, since the man climbed the hill at the rate (velocity) of 2 km/hr, so in an hour he can go to 2km. But it's given in the question that he climbed the hill in half an hour, that is 1 hr ---> 2 km 0.5 hr ----> (0.5*2 km) = (1 km)		
31	4			
32	3			http://stuffyoulook.blogspot.com/2013/09/step-to-step-animation-process.html
33	2			
34	1	• A photography with low shutter speed results in creating a more pronounced motion blur effect. notice that the remaining backgrounds (trees and wall) looks clean; only the part of movement or motion parts will appear blur. • If the shutter speed is high, the camera can catch even minute details of motion/movement, like you can even capture details of water splashing. • when there is a camera shake, all the parts of the picture should be blur. • If there is a mistake in saving, then the photo could not be opened at all.		http://stuffyoulook.blogspot.com/2013/10/photography-for-CEED-Part-A.html
35	2	Shadow lengths are directly proportional to the heights. so apply 25/40 = 15/x; x thus equals to 24		
36	2			
37	3	For this question, as the string makes 5 total turns in 15 cm length; assume only one turn in 3 cm length; now you have to imagine that the cylinder of 3 cm length is un-folded in flat surface; thus forming a rectangle of width equal to circle circumference (4 cm) and breadth equal to 3 cm. Notice that the string will assume a straight line if unfolded in flat surface. Now apply Pythagoras theorem to find the hypotenuse, which will turn out as 5. For the total length of cylinder, multiply this by 5 times; thus counting to 25		
38	12	If 'L' is the length of the pole. one half of the pole is burried in soil; so L/2 part is left, of which one third of the pole is submerged in water and thus the remaining part is (L/2-L/3) which assumes L/6, as given in question this should equal to 2 m. Thus		

		L/6 = 2 L = 12 m		
39	4	Check engineering isometric drawing examples for such questions		
40	1			
41	2	Up to last year, Citreon manufacturer doesn't have a plant. But recently they are setting a plant in Gujarat.		
42	2			
43	1			
44	B			
45	3			
46	4			
47	3	Frankly speaking, the cupboard on the room is not in perspective, but as per the options given, windows on the back wall should have projections on the wall (similar to window on side wall) which is missing here.		
48	3	Trick is to count the number of missing cubes, here it is 15 and deduct the quantity form the total number of cubes (4x4x4=64 cubes); thus summing to 49		http://stuffyoulook.blogspot.com/2013/09/working-with-geometric-objects.html
49	2	Front - 6 surfaces right - 6 surfaces back - 4 surfaces left - 2 surfaces top - 4 surfaces bottom - 1 surface		
50	3	the remaining pictures are a little bit confusing.		